

Homework 4, Due February 14, 2025

1. Refer to Problem 1 in Homework 2.

A class consists of $n=24$ students. The average BMI for the class is 32.46, and the standard deviation is 6.92. Six new students have joined the class. Their BMIs are [23.5, 30.5, 42.3, 34.5, 49.5, 44.2]. Write Python code and calculate the sample mean and standard deviation for the 30 students using the recursive formula.

2. The probability that a student owns a microwave oven is 0.75, and that a student owns a TV is 0.25. The probability that a student owns both a microwave and a TV is 0.16. Find the probability that a student owns a microwave or a TV, but not both.
3. Five cards are drawn from a standard deck of cards without replacement. Find the probability of getting
 - a. All red cards
 - b. All diamonds
 - c. All aces
4. Suppose a person is randomly selected from a population of 1000 people with the distribution given below in the table.

Age	Disease Status				Totals
	None	Mild	Moderate	Severe	
18 – 40	213	51	33	23	320
Over 40	430	121	98	31	680
Totals	643	172	131	54	1,000

Find the probabilities of the following events that the person is

- a. $P(\text{Over 40})$
- b. $P(\text{Mild and Over 40})$
- c. $P(\text{Mild or Over 40})$
- d. $P(\text{not Mild})$
- e. $P(\text{Mild} \mid \text{Over 40})$

The following three problems are from the textbook.

5. Page 81, Problem 3.8
6. Page 82, Problem 3.19
7. Page 83, Problem 3.26

Warning: independence can mislead	64
Conditional probability: lottery example	69
Intuitions about conditional probability are likely wrong; be careful	73

3.6.5 Be Able to

- Write out a set of outcomes for an experiment.
- Construct an event space.
- Compute the probabilities of outcomes and events.
- Determine when events are independent.
- Compute the probabilities of outcomes by counting events, when the count is straightforward.
- Compute a conditional probability.

Problems

Outcomes

- 3.1** You roll a four sided die. What is the space of outcomes?
- 3.2** King Lear decides to allocate three provinces (1, 2, and 3) to his daughters (Goneril, Regan and Cordelia—read the book) at random. Each gets one province. What is the space of outcomes?
- 3.3** You randomly wave a flyswatter at a fly. What is the space of outcomes?
- 3.4** You read the book, so you know that King Lear had family problems. As a result, he decides to allocate two provinces to one daughter, one province to another daughter, and no provinces to the third. Because he’s a bad problem solver, he does so at random. What is the space of outcomes?

The Probability of an Outcome

- 3.5** You roll a fair four sided die. What is the probability of getting a 3?
- 3.6** You shuffle a standard deck of playing cards and draw a card. What is the probability that this is the king of hearts?
- 3.7** A roulette wheel has 36 slots numbered 1–36. Of these slots, the odd numbers are red and the even numbers are black. There are two slots numbered zero, which are green. The croupier spins the wheel, and throws a ball onto the surface; the ball bounces around and ends up in a slot (which is chosen fairly and at random). What is the probability the ball ends up in slot 2?

Events

- 3.8** At a particular University, $\frac{1}{2}$ of the students drink alcohol and $\frac{1}{3}$ of the students smoke cigarettes.
- (a) What is the largest possible fraction of students who do neither?
- (b) It turns out that, in fact, $\frac{1}{3}$ of the students do neither. What fraction of the students does both?

Computing Probabilities by Counting Outcomes

3.9 Assume each outcome in Ω has the same probability. In this case, show

$$P(\mathcal{E}) = \frac{\text{Number of outcomes in } \mathcal{E}}{\text{Total number of outcomes in } \Omega}$$

3.10 You roll a fair four sided die, and then a fair six sided die. You add the numbers on the two dice. What is the probability the result is even?

3.11 You roll a fair 20 sided die. What is the probability of getting an even number?

3.12 You roll a fair five sided die. What is the probability of getting an even number?

3.13 *I am indebted to Amin Sadeghi for this exercise.* You must sort four balls into two buckets. There are two white, one red and one green ball.

- (a) For each ball, you choose a bucket independently and at random, with probability $\frac{1}{2}$. Show that the probability each bucket has a colored ball in it is $\frac{1}{2}$.
- (b) You now choose to sort these balls in such a way that each bucket has two balls in it. You can do so by generating a permutation of the balls uniformly and at random, then placing the first two balls in the first bucket and the second two balls in the second bucket. Show that there are 16 permutations where there is one colored ball in each bucket.
- (c) Use the results of the previous step to show that, using the sorting procedure of that step, the probability of having a colored ball in each bucket is $\frac{2}{3}$.
- (d) Why do the two sorting procedures give such different outcomes?

The Probability of Events

3.14 You flip a fair coin three times. What is the probability of seeing HTH? (i.e. Heads, then Tails, then Heads)

3.15 You shuffle a standard deck of playing cards and draw a card.

- (a) What is the probability that this is a king?
- (b) What is the probability that this is a heart?
- (c) What is the probability that this is a red card (i.e. a heart or a diamond)?

3.16 A roulette wheel has 36 slots numbered 1–36. Of these slots, the odd numbers are red and the even numbers are black. There are two slots numbered zero, which are green. The croupier spins the wheel, and throws a ball onto the surface; the ball bounces around and ends up in a slot (which is chosen fairly and at random).

- (a) What is the probability the ball ends up in a green slot?
- (b) What is the probability the ball ends up in a red slot with an even number?
- (c) What is the probability the ball ends up in a red slot with a number divisible by 7?

3.17 You flip a fair coin three times. What is the probability of seeing two heads and one tail?

3.18 You remove the king of hearts from a standard deck of cards, then shuffle it and draw a card.

- (a) What is the probability this card is a king?
- (b) What is the probability this card is a heart?

3.19 You shuffle a standard deck of cards, then draw four cards.

- (a) What is the probability all four are the same suit?
- (b) What is the probability all four are red?
- (c) What is the probability each has a different suit?

3.20 You roll three fair six-sided dice and add the numbers. What is the probability the result is even?

3.21 You roll three fair six-sided dice and add the numbers. What is the probability the result is even *and* not divisible by 20?

3.22 You shuffle a standard deck of cards, then draw seven cards. What is the probability that you see no aces?

3.23 Show that $P(\mathcal{A} - (\mathcal{B} \cup \mathcal{C})) = P(\mathcal{A}) - P(\mathcal{A} \cap \mathcal{B}) - P(\mathcal{A} \cap \mathcal{C}) + P(\mathcal{A} \cap \mathcal{B} \cap \mathcal{C})$.

3.24 You draw a single card from a standard 52 card deck. What is the probability that it is red?

3.25 You remove all heart cards from a standard 52 card deck, then draw a single card from the result.

- (a) What is the probability that the card you draw is a red king?
- (b) What is the probability that the card you draw is a spade?

Permutations and Combinations

3.26 You shuffle a standard deck of playing cards, and deal a hand of 10 cards. With what probability does this hand have five red cards?

3.27 Magic the Gathering is a popular card game. Cards can be land cards, or other cards. We consider a game with two players. Each player has a deck of 40 cards. Each player shuffles their deck, then deals seven cards, called their *hand*.

- (a) Assume that player one has 10 land cards in their deck and player two has 20. With what probability will *each* player have four lands in their hand?
- (b) Assume that player one has 10 land cards in their deck and player two has 20. With what probability will player one have two lands and player two have three lands in hand?
- (c) Assume that player one has 10 land cards in their deck and player two has 20. With what probability will player two have more lands in hand than player one?

3.28 The previous exercise divided Magic the Gathering cards into lands vs. other. We now recognize four kinds of cards: land, spell, creature and artifact. We consider a game with two players. Each player has a deck of 40 cards. Each player shuffles their deck, then deals seven cards, called their *hand*.

- (a) Assume that player one has 10 land cards, 10 spell cards, 10 creature cards and 10 artifact cards in their deck. With what probability will player one have at least one of each kind of card in hand?
- (b) Assume that player two has 20 land cards, 5 spell cards, 7 creature cards and 8 artifact cards in their deck. With what probability will player two have at least one of each kind of card in hand?
- (c) Assume that player one has 10 land cards, 10 spell cards, 10 creature cards and 10 artifact cards in their deck; and player two has 20 land cards, 5 spell cards, 7 creature cards and 8 artifact cards in their deck. With what probability will at least one of the players have at least one of each kind card in hand?

3.29 You take a standard deck of 52 playing cards and shuffle it. Compute the probability that, in the shuffled deck, there is at least one pair of cards following one another in increasing order (i.e. a 2 followed by a 3, or a 3 followed by a 4, etc.). This isn't particularly easy, but the probability is higher than most people realize; you can surprise your friends and make money with this information.